English-to-Khmer Phonetic Transliteration: Models, Data and Human Acceptability Evaluation

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Abstract

We study a narrow task: given one English word or name, write in Khmer script how it sounds when read in English. Two systems are released as research previews: a 2.6M-parameter character-level Transformer trained from scratch, and a LoRA adapter for ByT5-small. Both are trained on English–Khmer pairs mined from bracketed web text, the adapter additionally on spellings attested in the ALT Parallel Corpus; no LLM-generated labels are used in either released model. Early experiments scored models by exact agreement with single mined or curated reference spellings. An exploratory blinded review found that such agreement both rejects acceptable alternatives and accepts unwanted spellings, so the final evaluation uses blinded acceptable/not acceptable/uncertain judgments by a single project-affiliated judge on prospectively frozen samples of 120 word families. On two such samples the released adapter's spelling was judged acceptable for 64.2% and 75.0% of words. A longer training schedule and pronunciation-guided training each had the same observed acceptability as the released adapter on their own sample (differences 0.0 pp [−8.3, +8.3] and 0.0 pp [−7.5, +7.5]): no difference was detected, and the intervals permit gains or losses. Adding English pronunciation input produced no detectable change in reference agreement in four experiments, although the intervals do not exclude gains of a few points. Irregularly spelled words remain the weakest group, and in a small diagnostic, alternative beam candidates often contained an acceptable spelling that greedy decoding missed.

1 Introduction and task

A Khmer text-to-speech front-end receives English words inside Khmer text—names, product terms, loanwords—and needs them in Khmer script before a Khmer-only speech model can read them. This report documents a transliterator for that step. Its input is one English word or name; its output is a Khmer-script spelling that a Khmer reader would pronounce close to the word's *English* pronunciation. For example, the released adapter writes *coffee* as ខូហ្វី and *london* as ឡុនដុន.

The task differs from three neighbouring ones. It is not *translation*: meaning is ignored. It is not *romanised-Khmer conversion*, where Latin letters encode a Khmer word (*sok sabay*). And it is not reproducing *established loan spellings* regardless of pronunciation: the usual Khmer form of *coffee* is កាហ្វេ, which does not follow the English pronunciation; the task's target does.

Three properties make the task hard to measure. First, English spelling underdetermines pronunciation (*record*, *live*, irregular names), so a target requires a declared reading; this project uses the first General American reading in CMUdict [1]. Second, Khmer adaptation is many-to-one and one-to-many: the script has two consonant series, inherent vowels, subscript consonants and series-shifting diacritics (as in ម៉ and ប៊), so several spellings can render the same sound acceptably. Third, the available supervision is web text written by many people under different conventions.

The contributions are proportionate to a small project: (i) two implemented and released systems with a common command-line and Python interface; (ii) a documented data pipeline, including the repair of a selection error that silently removed dictionary-word targets; (iii) a set of controlled comparisons of pronunciation input, data, schedule and decoding, each registered before its results were seen; and (iv) a human-acceptability evaluation protocol with frozen inputs, blinded spellings and explicit labels. No new architecture or general benchmark is claimed.

2 Related work

Grapheme, phoneme and hybrid transliteration. Oh and Choi [12] distinguish grapheme-based, phoneme-based and hybrid models for English–Korean and English–Japanese transliteration and report gains from combining source graphemes and phonemes dynamically. Ngo et al. [11] model target-language phonology explicitly for low-resource transliteration into Vietnamese, motivated by phonologically invalid outputs. These works motivate the project's *hypothesis* that supplying CMUdict phones would help; whether it does for Khmer with small mined data is what Section 5.1 tests.

Byte-level pretrained models. ByT5 [16] operates on UTF-8 bytes and is reported to be more robust

to noise and stronger on spelling- and pronunciation-sensitive tasks than token-based counterparts. Kirov et al. [9] fine-tune ByT5 for single-word transliteration of romanised South Asian languages and find its fine-tuning sensitive to noisy training items: removing the 10% least representative pairs helped, while non-pretrained models did not benefit. The adapter released here uses ByT5-small with LoRA [7]. Khmer characters are three UTF-8 bytes each, a detail that matters for adaptation (Section 3.2).

Evaluation with multiple references. The NEWS 2018 shared task [4] scores transliteration against reference *lists*, where any listed variant counts (ACC), alongside mean F-score, MRR and MAP. This project has one mined spelling per item for most of its history, so only exact agreement and character error rate are computed against it; the human evaluation instead judges each produced spelling, which admits any number of acceptable variants.

Candidate generation and rescoring. Grundkiewicz and Heafield [6] apply NMT techniques to named-entity transliteration, including ensembling and rescoring n-best lists with right-to-left models. Rescoring can only help when acceptable spellings are already in the candidate list; Section 5.6 measures that presence for the released adapter. No reranker is built here.

3 Data and models

3.1 Mined pairs and the selection repair

Training pairs come from Khmer web text in which a Khmer spelling is followed by the English word in brackets, e.g. "ចនសុន (Johnson)", mined from the Khmer subset of FineWeb-2 [8, 13] and a Khmer Wikipedia dump of 2026-10-01 [15]. A pair is kept when the English word is in CMUdict, passes a noise filter, does not look like romanised Khmer, and belongs to no word family used for evaluation. A *family* groups an English word with its plurals and other inflections; exclusion and bootstrap resampling both operate on families so that *band* and *bands* cannot fall on opposite sides of a split.

The original miner labelled a pair `khmer_like`, and dropped it, if its Latin side was a known romanised Khmer word *or* its Khmer side was a known Khmer word. The second condition removed exactly the pairs whose Khmer spelling is in the Google Khmer lexicon [5], including established loans; a later diagnostic found 0 of 10,031 training targets covered by exact lexicon lookup, by construction. Separating the two conditions recovered 11,377 rows; 1,507 romanised-Latin rows stayed excluded.

Consonant screening and its limits. To check that a mined spelling plausibly renders the English word, a rule-based reader maps Khmer consonants to sound classes and compares them with the CMUdict reading. On 30,980 held-out lexicon headwords its consonant classes agreed with the lexicon's own transcriptions for 88.6%. The screen does not read vowels at all. After the repair, 2,967 dictionary-assessable, consonant-screened pairs remained. A later audit recovered 1,292 pairs that the screen had rejected because of documented weaknesses (for example doubled letters and glides), and added 1,217 names and loanwords whose Khmer spelling occurs verbatim in the human translations of the ALT Parallel Corpus [10, 14], passed through the same screen. None of these spellings was reviewed by a person, and vowel quality is unassessed throughout.

Synthetic labels (historical only). One data-expansion experiment added 384 rows for 193 words whose spellings were suggested by Google's AI Mode. They were used only in historical runs. The released adapter's training file contains none (verified row by row), and the released small model never used them.

3.2 The two released models

Table 1 summarises the releases.

Small character-level Transformer. An encoder–decoder with $d_{\text{model}} = 192$, three encoder and three decoder layers, four heads and feed-forward width 512 (2.6M parameters), trained from scratch on 9,735 mined pairs (296 more held out for validation loss) with AdamW, learning rate 10^{-3}, cosine decay and label smoothing 0.1. The epoch (27 of 40) was selected on 86 development items; final scores come from separate test items. Inference runs in NumPy with beam search of width 4. The released weights are the letters-only seed-0 run (`letters_s0`); the file on the Hub is byte-identical to the recorded checkpoint (sha256 `8901050b…`).

ByT5-small with a LoRA adapter. The base `google/byt5-small` (revision `68377bdc`) is frozen. LoRA matrices of rank 16 and $\alpha = 32$ are added to every attention projection (q, k, v, o) and gated feed-forward layer (wi_0, wi_1, wo): 256 tensors, 6,258,688 parameters. Training used 4,764 pairs over 3,585 words—2,255 screened mined pairs, 1,292 recovered pairs and 1,217 ALT-attested pairs—with AdamW at 10^{-3}, 100 warm-up steps then linear decay, batch 32 and 3,000 updates (4.6 hours on six CPU threads, one seed). Checkpoints were selected by greedy exact agreement on 253 development items, then CER. The selected checkpoint was the last one evaluated; that alone does not show the model was under-trained, and a twice-longer schedule did not change acceptability (Section 5.3).

An earlier ByT5 attempt with attention-only adapters and 1,000 updates produced nearly the same output for every input (ហ្ន for 204 of 228 evaluation

words). The audit found no pipeline defect explaining the collapse: two of the three UTF-8 bytes of a Khmer character are almost free to predict, so a falling per-byte loss coexisted with an input-independent letter model. Adding feed-forward adapters and training longer produced an input-dependent model; which change mattered was not isolated.

The adapter is stored in a project format (`adapter.safetensors` with keys ending in `.A`/`.B` plus a JSON configuration), not as a PEFT checkpoint. It loads only through the package's `khmer_translit.byt5` module, which wraps the base model's linear layers and refuses a base whose revision or weight hash differs from the pinned one. Single answers use greedy decoding; candidate lists use beam search with `early_stopping="never"` and no length penalty (Section 5.4).

Data separation. For each experiment, training, validation, checkpoint selection and evaluation used disjoint families. The small model's final references were opened once, after all seven of its comparison runs were frozen. The adapter experiments share a *historical* development set (253 items) and evaluation set (228 items, 222 families) that were examined repeatedly across eleven experiments; scores on them are exploratory. The human tests (Section 4) used fresh families excluded from all training files. The two released models were trained on different data and were never evaluated on a common, clean test set. The historical evaluation set overlaps the small model's training words: 110 of its 228 primary evaluation words occur in that training file (122 of 245 words in the full evaluation partition). It therefore gives no clean unseen-word comparison between the released models, and no head-to-head comparison is reported.

4 Evaluation methodology

Reference agreement. Early experiments scored the top output against one reference spelling per item: exact agreement, averaged over families, and character error rate (CER). For the small model the references were a curated primary list of 165 items in 162 families, spellings checked by the author, 82 of whose references were chosen from an earlier model's candidates (94 of 191 items across the primary and a 26-item flagged track); and an exploratory list of 400 LLM-suggested, spot-checked spellings. For the adapters they were the mined spellings themselves. Agreement is not phonetic accuracy: one acceptable spelling per item cannot admit legitimate variants, and an unreviewed reference can itself be wrong.

Intervals. Paired differences between two systems use a family-level bootstrap with 10,000 resamples. These intervals are *conditional* on the trained checkpoints: they describe item sampling, not training-seed variation, and exclude the extra uncertainty of checkpoint selection. Where three paired seeds exist, their per-seed differences are reported as well. Intervals for differences are not intervals for either model's own rate, and a difference whose interval includes zero is not evidence of equivalence.

Human acceptability. A spelling is *acceptable* if a Khmer reader would pronounce it close to the declared General American reading, *not acceptable* otherwise, and *uncertain* when the judge cannot decide; the policy file was fixed before the reviews and unchanged across them. The judge is the project's author—a single, project-affiliated judge. Spellings were shown without model identity, rank or score, with labels in seeded random order, and identical spellings from two systems were shown once. In the two prospective tests and for the new candidate-coverage judgments, the judge answered each spelling with an explicit click on a local review page; unanswered spellings count as *missing*, never as rejections, and an unjudged spelling stays unassessed. The earlier development review was answered in chat instead and transcribed under three interpretation rules the judge confirmed, including rules for unnamed and skipped candidates (Appendix C).

Two sampling designs were used and are never pooled. A development review of 60 words used a representative sample of 40 words and an enriched sample of 20 words where the compared systems disagreed, reported separately; it was a post hoc review of existing models. Two prospective tests then each drew 120 fresh families—40 ordinary words, 40 irregular spellings and 40 names—frozen before the candidate model was trained, with all predictions frozen before review. "Confirmed acceptable" counts uncertain as not acceptable; sensitivity analyses recount uncertain both ways. These judgments are useful task-specific evidence of acceptability for this judge and policy; they are not independent, and the 40/40/40 design does not estimate performance on everyday text.

5 Results

Table 2 lists the principal comparisons; Table 3 gives the blinded tests.

5.1 Did pronunciation input help?

No experiment detected a benefit. In the small model's registered comparison, adding CMUdict phones next to the letters changed curated-reference agreement by +1.6 pp on average over three paired seeds (per seed +0.6, +1.5, +2.8), with interval [−1.6, +4.9]. The interval excludes the registered 5-point threshold and includes zero; with simulated power of only 0.22–0.50 for a true 5-point gain across the registered discordance assumptions, the result is inconclusive, not null. Repeating the comparison on dictionary-assessable, consonant-screened pairs and with a phones-only arm gave the same picture (Table 2); a phones-only model scored −8.0 pp against letters (one seed) but got different words right, and

Table 1: The two released systems. Latency and memory: Intel Core i7-8750H laptop, CPU only; the small model's latency is the range of three recorded runs.

	Small Transformer	ByT5-small + LoRA adapter
Repository	`ric912/en-km-phonetic-transliterator` (tag v0.2.0)	`ric912/en-km-phonetic-byt5-lora`
Parameters	2.6M, trained from scratch	300M frozen base + 6.26M adapter
Training pairs	9,735 mined (FineWeb-2, Wikipedia)	4,764 (mined, recovered, ALT-attested)
Runtime	NumPy only	PyTorch, Transformers, safetensors; base fetched separately (1.2 GB)
Decoding	beam 4	greedy; beam ≥ 4 for candidates
Load time	0.15–0.16 s	5.5–7.1 s
One word (median)	17–19 ms	554 ms (greedy), 820 ms (top-4)
Batched	13 ms per word (batches of 16)	14.5 words/s (greedy)
Peak memory	65 MiB	2019 MB
Weights licence	CC BY-NC-SA 4.0	CC BY-NC-SA 4.0 (base: Apache 2.0)

Table 2: Principal comparisons. Δ is in percentage points with a 95% paired family-bootstrap interval, conditional on the checkpoints. Reference agreement is exploratory except where marked.

Question	Comparison (experiment)	Δ [95%]	Evidence and limits
Pronunciation input	small model, letters + phones vs letters (E02)	+1.6 [−1.6, +4.9]	curated references, 162 families; 3 paired seeds, all positive; registered, inconclusive
	same, on consonant-screened pairs (E03)	+0.6 [−2.4, +3.8]	mined references; 3 seeds
	letters + phones vs letters (E04)	+2.2 [−1.5, +6.0]	mined references; 3 seeds
	pooled letters+phones candidates vs letters top-8 (E05)	+1.6 [−1.1, +4.5]	top-8 oracle, equal budget
	adapter with CMUdict reading vs released adapter (E13)	−1.3 [−4.8, +2.3]	mined references; several ingredients changed
ByT5 adaptation	corrected adapter vs mean of 3 small letters models (E07)	+5.8 [+0.05, +11.6]	registered beam-4; one adapter run
More data	+ recovered, ALT and synthetic rows vs screened mined pairs only (E08 vs E07)	+0.3 [−5.7, +6.5]	registered beam-4; greedy +2.6 [−2.4, +7.5]
	real data only vs E07, greedy (E11 vs E07)	+2.5 [−2.7, +7.7]	one run each
	real data only vs with synthetic rows (E11 vs E08)	−0.1 [−4.7, +4.6]	one run each
Longer training	6,000 vs 3,000 updates (E12 vs E11)	+0.7 [−4.1, +5.4]	one seed; blinded test in Table 3

pooling the two models' candidates did not beat a letters model's own top-8 list at equal budget.

Sensitivity diagnostics explain part of this. Supplying another word's phones changed the small model's top output for 33.0–44.7% of 94 words, depending on the seed, but the changed outputs moved toward the donor's consonants in only 2–7 cases per seed; most changes were vowel or diacritic edits, and letters-only controls, as intended, never changed. The pronunciation-guided adapter behaved similarly: removing the reading changed 7.1% of historical-development outputs and a wrong reading 7.9%. These diagnostics show weak dependence on the supplied phones in these models and data; they do not show that pronunciation information cannot help Khmer transliteration.

5.2 Did more real data help?

Recovering screened-out pairs and adding ALT-attested spellings roughly doubled the adapter's data. Against the first corrected adapter, the released real-data adapter's greedy agreement on the historical evaluation set was 52.3% vs 49.9% (+2.5 pp, [−2.7, +7.7]), and a version that also used the synthetic labels was indistinguishable (−0.1 pp). Reference agreement did not distinguish these models. The exploratory development review suggested a preference for the expanded real-data model: its greedy spelling was acceptable for 29 of 40 representative words against 24 for the first corrected adapter, and for 10 of 20 words in the disagreement-enriched sample, reported separately, against 5; the synthetic-label version scored 30 and 8. The review was post hoc and had one judge, so it is development evidence, not a registered test.

5.3 Did longer training help?

Doubling the schedule to 6,000 updates changed 65 of 120 test spellings. The two models had equal observed acceptability, 77 vs 77 of 120 (0.0 pp, [−8.3, +8.3]; 14 vs 14 discordant words, exact $p = 1.00$): no difference

was detected, and the interval permits gains or losses. Historical agreement moved by +0.7 pp. One seed was run.

5.4 How did decoding configuration affect results?

For the adapters, greedy decoding had beaten the registered beam-4 search. An inference-only audit traced this to `early_stopping=True`: at the byte level, junk hypotheses (an immediate end symbol, or one letter and an end symbol) finish within the first steps and fill the beam's finished set before real spellings end. Registered beam-4 kept only 3.60 distinct candidates per item for the synthetic-data adapter and often dropped the series-shifting diacritics (*mani* ម៉ានី became មានី). With `early_stopping="never"`, beam top-1 was about as good as greedy (+0.9 pp, [−1.4, +3.6]) and +5.9 pp [+1.4, +10.4] above the registered beam on the historical evaluation set. Earlier ByT5 comparisons that used the registered beam are therefore shifted; this report quotes greedy figures for adapters where they exist.

5.5 How does reference agreement differ from acceptability?

In the development review, the judge rated 109 distinct spellings. Of the 72 that differed from the mined reference, 28 (39%) were acceptable (for example *cameroon* ខេមរ៉ុន and *band* បេន); of the 37 that matched it, 8 (22%) were not acceptable (*band* ប៉ាន). Exact agreement can therefore both reject acceptable alternatives and accept unwanted spellings; it is not a one-directional underestimate. Figure 1 shows the two kinds of evidence side by side, on separate axes because they are measured on different items with different definitions.

5.6 Is candidate ranking a plausible remaining opportunity?

For 23 irregularly spelled development words (frozen before generation, none from test families), the judge rated every distinct candidate among the adapter's greedy output and four canonical beam candidates (all 93 distinct spellings judged: 90 by clicks on the review page and 3 reused from the chat-transcribed development review; earlier judgments were reused only for the identical word, reading and spelling). The greedy output was acceptable for 8 words and beam top-1 for 10; at least one of the four beam candidates was acceptable for 15. A perfect selector over those four would add 7 words relative to greedy and 5 relative to beam top-1; 8 words had no acceptable candidate. This oracle is an upper bound for a reranker on this sample, not achieved accuracy, and 23 hand-picked hard words cannot show whether ranking or generation is the larger limitation in general.

Pronunciation-guided training on the second sample. The second test evaluated a model that changed several things at once: CMUdict readings as input, the judge's accepted spellings of 42 difficult training families mixed into 10% of batches, replay of the original data, and initialisation from the released adapter. The models had equal observed acceptability (90 vs 90; 11 vs 11 discordant words; 0.0 pp [−7.5, +7.5]): no difference was detected, and the interval permits gains or losses. Because the ingredients were not varied separately, this result cannot attribute any effect, or its absence, to one of them. Across both samples, irregular spellings were the weakest group (14/40 and 20/40 for the released adapter).

6 Released systems and reproducibility

Both models are published under CC BY-NC-SA 4.0 for the weights [2, 3]; the `khmer-translit` package that runs them is under PolyForm Noncommercial 1.0.0 and its wheel is attached to the small model's repository. The small model needs only NumPy and answers in tens of milliseconds. The adapter needs PyTorch and Transformers and a separate download of the pinned base; its median single-word latency is 29–33 times the small model's and its peak memory 2019 MB, so it suits drafting rather than high-volume use on a CPU.

Before release the adapter export was verified to be bit-identical to the training checkpoint (sha256 `286292a4…`), and the packaged loader reproduced the frozen research predictions with 0 mismatches on 276 development and 0 on 245 evaluation words (greedy) and 0 mismatches in top-4 lists. A fresh, isolated CPU-only installation passed all 19 checks with network access blocked at the socket level, including refusal of empty, multi-word, non-Latin and over-long inputs and of corrupt or mismatched files. The published repositories were re-downloaded and the quickstart commands re-run. Both interfaces share one command (`khmer-translit predict`); exact usage is in each model card and in the report's companion README. Training data, judgments and references are not distributed.

7 Limitations and conclusion

Limitations. *Supervision* is mined and unreviewed; the consonant screen ignores vowels, which were the largest class of reference disagreement for every adapter (41–49% of mismatches). *References* cover one spelling per item, and much of the reference-based evidence comes from a historical evaluation set examined across many experiments. *Human evaluation* rests on one project-affiliated judge, samples of 120 words (each rate is uncertain by about ±7.7–8.6 points, normal approximation), a 40/40/40 design that over-represents hard words and names, and

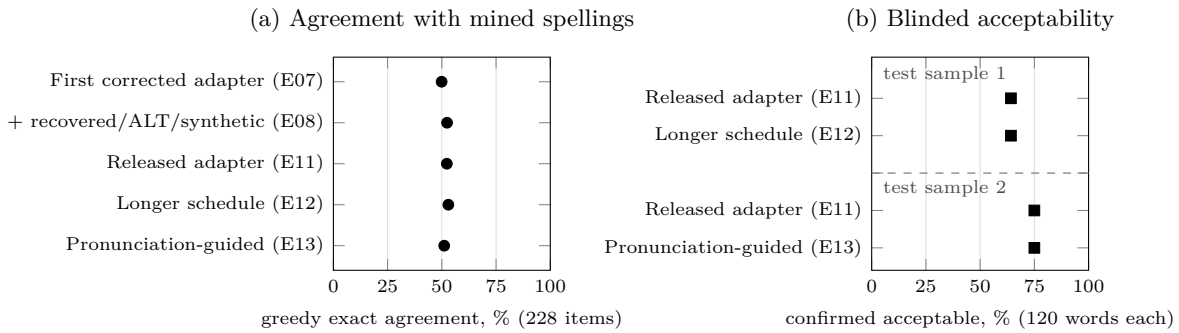


Figure 1: Two different measures, not a common benchmark. (a) Historical evaluation set, examined repeatedly, single unreviewed reference per item. (b) Two separate prospective samples; each compares its candidate with the released adapter on the same 120 words, and values are not comparable across samples. No uncertainty bars are drawn for individual rates; paired differences are in Table 3.

Table 3: Prospective blinded tests (one judge). Each sample compares a candidate with the released adapter on its own 120 fresh families; the two samples are different and are not pooled. Uncertain (Unc.) counts as not confirmed. Ord. = ordinary words, Irreg. = irregular spellings. Δ: candidate minus released adapter, percentage points, 95% paired family bootstrap.

Sample	System	Acceptable of 120	Unc.	Ord. of 40	Irreg. of 40	Names of 40	Δ [95%]
1 (E12)	Released adapter (E11)	77 (64.2%)	3	27	14	36	
	6,000 updates (E12)	77 (64.2%)	4	27	19	31	0.0 [−8.3, +8.3]
2 (E13)	Released adapter (E11)	90 (75.0%)	1	34	20	36	
	Pronunciation-guided (E13)	90 (75.0%)	1	32	21	37	0.0 [−7.5, +7.5]

one training seed per adapter. *Pronunciation policy* fixes one General American reading per word, and CMUdict coverage limits which words can be checked. *Permissions*: rights in the mined web pages remain with their authors; FineWeb-2 is distributed under ODC-By 1.0 and Common Crawl's terms [8], and whether Wikipedia's share-alike condition [15] extends to trained weights is legally unsettled. *Scope*: single words only; sentences, context and romanised Khmer are out of scope.

Conclusion. Two English-to-Khmer phonetic transliterators are documented and released with their evidence. Within this project's data and models, English pronunciation input, more training updates and pronunciation-guided training produced no detectable gain, and a decoding configuration error—not model quality—explained an apparent beam search deficit for byte-level models. The most informative measurement change was moving from single-reference agreement to blinded acceptability judgments; an exploratory development review suggested a preference between models that agreement did not distinguish. On two separate 120-word samples, the released adapter's spelling was judged acceptable for 64–75% of words, and for 35–50% of irregularly spelled words. The candidate diagnostic indicates that acceptable alternatives are often already generated, while 35% of the hard words had none; which of the two dominates on ordinary text remains unmeasured.

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A Experiment inventory

Table 4: All experiments, in order. "Ref." = agreement with single reference spellings (exploratory unless noted); "Human" = blinded judgments by the author. E02d and E11c are diagnostics attached to E02 and E11.

Id	Question	Outcome	Evidence
E01	Khmer-pronunciation side task in one model for names, romanised Khmer and English	+4.67 pts on English, missed its 5-point rule	Ref., registered
E02	Phones vs letters, small model (released: `letters_s0`)	+1.6 [−1.6, +4.9]; inconclusive; letters kept	Ref. (curated), registered
E02d	Do targets follow the phones; do models use them?	0 of 10,031 targets found by exact lexicon lookup (by construction); shallow sensitivity	diagnostics
E03	Selection repair; phones vs letters on screened pairs	11,377 rows recovered; +0.6 [−2.4, +3.8]	Ref., registered
E04	Letters vs phones-only vs both	both − letters +2.2 [−1.5, +6.0]	Ref., registered
E05	Pooling letters and phones candidates (inference only)	no clear advantage at equal budget	Ref. (oracle)
E06	ByT5-small + attention-only LoRA, 1,000 updates	input ignored; 0 of 228 eval items correct	Ref.
E07	Why E06 collapsed; corrected adapter	no pipeline defect explaining the collapse; +5.8 [+0.05, +11.6] vs small letters models	Ref., registered
E08	Data expansion (recovered, ALT, 384 synthetic rows)	+0.3 [−5.7, +6.5] vs E07	Ref.
E09	Why greedy beat beam (inference only)	`early_stopping=True` defect; canonical beam ≈ greedy	Ref.
E10	Word-level sampling + higher learning rate	development gate failed; not adopted	Ref.
E11	Real data only, no synthetic labels (released adapter)	+2.5 [−2.7, +7.7] vs E07; post hoc dev review 29/40 representative, 10/20 disagreement-enriched	Ref.; Human (dev, chat)
E12	6,000-update schedule	equal observed acceptability, 0.0 [−8.3, +8.3]	Human (test 1), registered
E13	Pronunciation-guided adapter	equal observed acceptability, 0.0 [−7.5, +7.5]	Human (test 2), registered
E11c	Candidate coverage, 23 irregular dev words	greedy 8, beam top-1 10, any of 4 15	Human (dev diagnostic)

B Data and evaluation pipeline

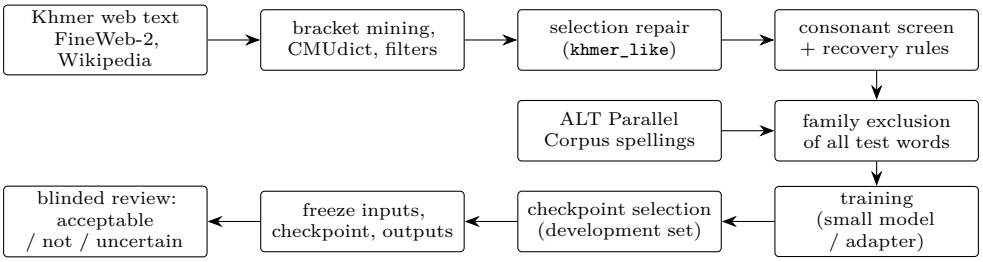


Figure 2: Data and evaluation pipeline of the released adapter (ALT spellings pass the same consonant screen). The small model used the mining and filter steps before the selection repair and was evaluated against reference lists rather than by blinded review.

C Notes on the evidence

- Empirical values are generated by `tools/extract.py` from the project's committed files at revision `d7e7d80`. Most are read from JSON results or derived from them. A few exist only in a committed report's text or table and are recorded as labelled transcriptions with file and section: the first test's per-group counts, the development review's reference cross-table, the number of changed test-1 spellings, the number of synthetic-label words, the development-set size of the small model and the number of difficult E13 families. `EVIDENCE_LEDGER.md` maps every value to its source. Architecture and recipe settings, sample-design constants, dates and bibliographic data are written literally.
- The development review was answered in chat. Its labels were transcribed under three rules the judge confirmed: candidates of an item not named in the reply are not acceptable; "ok" without a label means all of the item's candidates are acceptable; a skipped single-candidate item is acceptable. The judge then clarified the skipped multi-candidate items, and a second reply used "not sure" (all candidates uncertain) and "maybe C too" (C uncertain), after which the judge confirmed three items. No spelling was typed. The two prospective tests and the new coverage judgments were clicked on the review page. Pooled 60-word percentages in some project notes mix the review's two samples and are not used.
- The released small model's published latency (17/30 ms) is from the first of three recorded measurement runs; later runs gave 19/34 and 19/38 ms (median/95th percentile); Table 1 gives the range. Parity checks used all 276 development rows and 245 evaluation rows, including items outside the primary scoring sets (253 and 228 items).